Lección 12

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Plan de trabajo

Formulación Lagrangiana

—Aplicaciones:

*Observadores no inerciales

*Movimiento de Sólido rígido

► Nuestro Syllabus:

Nuestro Syllabus:

FIS-310:

UNIDAD 2. Formulación Lagrangiana

2.1. Variables Generalizadas

2.2. Clasificación de los Sistemas Mecánicos

2.3. Elementos de Cálculo Variacional

2.4. Ecuaciones de Lagrange

2.5. Aplicaciones: movimiento en un campo central, observadores no inerciales

FIS-345:

Unidad I: Principios Variacionales, Lagrangianos y Hamiltonianos

- Principio de Hamilton y Lagrangianos
- Simetrías y Leyes de Conservación (Teorema de Noether)

Unidad II: Algunas Aplicaciones

- Movimiento en campo de Fuerzas Centrales
- Movimiento de Sólido Rígido

Motion of rigid body

- Six degrees of freedom: 3 translations + 3 rotations
- In this course we'll see that sometimes consideration is simpler when we work in the non-inertial body's reference frame
  –In that frame body's shape/distribution of mass is constant

Rotation matrix R:

–Relates components of vectors in two frames (A, B):

$$v_i^{(\mathrm{A})} = R_{ij} v_j^{(B)}, \qquad v_i^{(B)} = R_{ij}^{-1} v_j^{(A)}, \qquad i, j = 1, 2, 3$$

for example in lab frame and body's rest frame

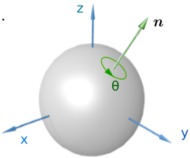
–Orthogonal, $\in SO(3)$ $[R^T R = 1]$

–Eigenvalues: 1, $e^{+i\theta}, e^{-i\theta}$ [Euler's theorem]

–Rotation axis: axis in direction of eigenvector n which has eigenvalue 1

–Rotation angle: value of θ

–For any rotation matrix $\mathrm{Tr}(R) = 1 + 2\cos\theta$



Explicit expression for rotation matrix:

$$R_{ab} = n_a n_b + \cos\theta\, (\delta_{ab} - n_a n_b) + \varepsilon_{akb} n^k \sin\theta$$

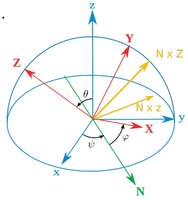
Alternative parametrization of rotation matrix

Euler angles

Represent any rotation as superposition of "elementary" rotations

$$R(\psi, \theta, \varphi) = R_Z(\varphi) R_X(\theta) R_Z(\psi)$$

- We'll use ZXZ convention for definiteness
- Can use any other combination in which two adjacent letters do not coincide: XYZ, ZYZ, ... (all are related)



$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Previous discussions: rotation vector $\boldsymbol{\theta}$ (or Euler angles ψ, θ, ϕ) are constant parameters.

Dynamical problems: ($\boldsymbol{\theta}$ or ψ, θ, ϕ) become functions of time. Even direction of $\boldsymbol{\theta}$ is NOT constant, depends on time

—Coordinates $r_b^{(\mathrm{body})}$ of any fixed point in the body's rest frame are related to its coordinates in the lab frame as

$$r_a^{(\mathrm{lab})}(t) = R_{ab}\left(\psi(t),\, \theta(t),\, \varphi(t)\right) r_b^{(\mathrm{body})} + X_a(t), \quad a = 1, 2, 3 \;\; (a = x, y, z)$$

where $X_a(t)$ are coordinates of in the body's center-of-mass ...

*The density $\rho = \rho\left(\boldsymbol{r}^{(\mathrm{body})}\right)$ does not depend on time in the body's rest frame

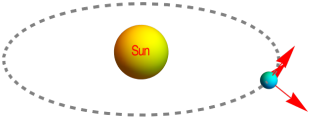
** ... but depends on time when we write it in terms of $\boldsymbol{r}^{(\mathrm{lab})}$

—In mechanical problems $\psi(t)$, $\theta(t)$, $\phi(t)$, $X_a(t)$ in general should be treated as degrees of freedom (rotations & translations)

*Time dependence should be fixed from (Euler-Lagrange) equations of motion & initial conditions

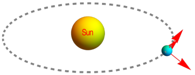
Control question

Write out the transformation which relates the components of vectors in the reference frames of the Earth and the Sun. For the rotation matrix R use parametrization in terms of the Euler angles.



Kinetic energy of the rotating body

Kinetic energy of rigid body



The rigid body (e.g. the Earth) rotates with constant angular velocity $\boldsymbol{\Omega}$ around axis $\boldsymbol{n}$ passing through its center of mass (direction of vector $\boldsymbol{\Omega}$ coincides with rotation axis, and $|\boldsymbol{\Omega}|$ is the absolute value of angular velocity). Also, the center of mass of the body moves with velocity $\boldsymbol{V}$ in the lab frame. Find the kinetic energy of the rigid body. Demonstrate that kinetic energy may be separated into 2 pieces,

$$T = T_{\rm CM} + T_{\rm rotation}$$

- Assume that in the Earth rest frame the position of the element of mass $dm = \rho dV$ is given by $\boldsymbol{r}_0 = \mathrm{const}$ w.r.t. the center of the Earth

  In the Sun's rest frame:
  ... its position is given by $\boldsymbol{r}(t) = R\left(\hat{\theta}(t)\right)\boldsymbol{r}_0 + \boldsymbol{R}_{\rm CM}(t)$
  ...At the moment $t + dt$, the position vector $\boldsymbol{r}(t + dt)$ slightly displaced with respect $\boldsymbol{r}(t)$:
  *For infinitesimal rotation angles $\delta\boldsymbol{\theta} \ll 1$ we may write the expressions for displacements of each element dm_i and corresponding velocities as

$$\delta\boldsymbol{r} = \boldsymbol{r}(t + dt) - \boldsymbol{r}(t) = \delta\boldsymbol{\theta}(t) \times (\boldsymbol{r} - \boldsymbol{R}_{\rm CM}) + \delta\boldsymbol{R}_{\rm CM}(t),$$
$$\boldsymbol{v} \equiv \delta\dot{\boldsymbol{r}} = \dot{\boldsymbol{\theta}} \times (\boldsymbol{r} - \boldsymbol{R}_{\rm CM}) + \boldsymbol{V} = \boldsymbol{\Omega} \times (\boldsymbol{r} - \boldsymbol{R}_{\rm CM}) + \boldsymbol{V}$$

Tensor of inertia

$$\boldsymbol{v} = \boldsymbol{\Omega} \times (\boldsymbol{r} - \boldsymbol{R}_{\mathrm{CM}}) + \boldsymbol{V}, \qquad \boldsymbol{R}_{\mathrm{CM}} = \frac{\int \rho(\boldsymbol{r})\, \boldsymbol{r}\, dV}{M}, \quad M \equiv \int \rho(\boldsymbol{r})\ dV$$

Kinetic energy:

$$T = \frac{1}{2}\int d^3r\, \rho(\boldsymbol{r}) \boldsymbol{v}^2(\boldsymbol{r}, t) =$$

$$\frac{1}{2} M \boldsymbol{V}^2 + \frac{1}{2}\int d^3r\, \rho(r)\, (\boldsymbol{\Omega} \times (\boldsymbol{r} - \boldsymbol{R}_{\mathrm{CM}}))^2 + \boldsymbol{V} \cdot \left(\boldsymbol{\Omega} \times \underbrace{\int d^3r\, \rho(r)\, (\boldsymbol{r} - \boldsymbol{R}_{\mathrm{CM}})}_{0} \right)$$

$$= \underbrace{\frac{1}{2} M \boldsymbol{V}^2}_{\text{CM motion}} + \frac{1}{2}\Omega_j \Omega_k \underbrace{\int dV\, \rho(r)\, \left(\delta_{jk} \mathfrak{r}^2 - \mathfrak{r}_j \mathfrak{r}_k\right)}_{I_{jk}}, \qquad \boldsymbol{\mathfrak{r}} \equiv \boldsymbol{r} - \boldsymbol{R}_{\mathrm{CM}}$$

I_{jk} is called the tensor of inertia
−We'll derive the red-colored expression in the next slide
−Kinetic energy indeed separates into 2 pieces, CM motion and rotation.

$$T = T_{\mathrm{CM}} + T_{\mathrm{rotation}}$$

Technical remark on vector algebra

Lemma

$$[\boldsymbol{A} \times \boldsymbol{B}] \cdot [\boldsymbol{C} \times \boldsymbol{D}] = (A \cdot C)(B \cdot D) - (A \cdot D)(B \cdot C)$$

If we substitute $\boldsymbol{A}$, $\boldsymbol{C} \to \boldsymbol{\Omega}$ and $\boldsymbol{B}, \boldsymbol{D} \to \boldsymbol{r}$, we'll get

$$[\boldsymbol{\Omega} \times \boldsymbol{r}]^2 = \boldsymbol{\Omega}^2 \boldsymbol{r}^2 - (\boldsymbol{\Omega} \cdot \boldsymbol{r})^2$$

Proof:

$$[\boldsymbol{A} \times \boldsymbol{B}] \cdot \boldsymbol{Z} = \boldsymbol{A} \cdot [\boldsymbol{B} \times \boldsymbol{Z}]$$

$$\boldsymbol{Z} \to \boldsymbol{C} \times \boldsymbol{D}$$

$$\boldsymbol{B} \times [\boldsymbol{C} \times \boldsymbol{D}] = \boldsymbol{C}\,(B \cdot D) - \boldsymbol{D}\,(B \cdot C)$$

$$\Downarrow$$

$$[\boldsymbol{A} \times \boldsymbol{B}] \cdot [\boldsymbol{C} \times \boldsymbol{D}] = (A \cdot C)(B \cdot D) - (A \cdot D)(B \cdot C)$$

Technical remark on vector algebra

$$[\boldsymbol{\Omega} \times \boldsymbol{r}]^2 = \boldsymbol{\Omega}^2 \boldsymbol{r}^2 - (\boldsymbol{\Omega} \cdot \boldsymbol{r})^2$$

In components (Cartesian coordinates):

$$\begin{aligned}
\Omega^2 &= \sum_{ij} \Omega_i \Omega_j \delta_{ij} \\
(\boldsymbol{\Omega} \cdot \boldsymbol{r}) &= \sum_i \Omega_i r_i, \\
(\boldsymbol{\Omega} \cdot \boldsymbol{r})^2 &= (\boldsymbol{\Omega} \cdot \boldsymbol{r})\,(\boldsymbol{\Omega} \cdot \boldsymbol{r}) = \sum_i \Omega_i r_i \sum_j \Omega_j r_j = \sum_{ij} \Omega_i \Omega_j r_i r_j
\end{aligned}$$

$$\Rightarrow [\boldsymbol{\Omega} \times \boldsymbol{r}]^2 = \sum_{ij} \Omega_i \Omega_j \left(\delta_{ij} r^2 - r_i r_j\right)$$

Properties of tensor of inertia

$$I_{jk} = \int d^3r\, \rho(\boldsymbol{r}, t) \left(\delta_{jk}\mathfrak{r}^2 - \mathfrak{r}_j \mathfrak{r}_k\right)$$

Some basic properties of the tensor of inertia I_{jk}:
−Symmetric 3×3 matrix
−Positively defined
−Depends on orientation (via $\rho(\boldsymbol{r}, t)$), is constant in body rest frame
−Explicit expression for components:

$$I_{jk} = \int dV\, \rho(\boldsymbol{r},\, t) \begin{pmatrix} y^2 + z^2 & -xy & -xz \\ -xy & x^2 + z^2 & -yz \\ -xz & -yz & x^2 + y^2 \end{pmatrix}$$

*For rotation around any axis $\boldsymbol{n}$, its <u>moment of inertia I_n</u> is defined by

$$I_n = \sum_{jk} I_{jk} n_j n_k = \vec{n} \cdot \hat{I} \cdot \vec{n}, \qquad I_{ij}\Omega_i\Omega_j \to I_n\Omega^2$$

(useful when rotation axis does not change its direction)
*Diagonal components (red) coincide with moment of inertia around axes x, y, z respectively:

$$I_{xx} \quad \leftrightarrow I_x, \qquad I_{yy} \leftrightarrow I_y, \qquad I_{zz} \leftrightarrow I_z$$

Tensor of inertia: transformation of components

$$I_{jk} = \int dV\, \rho(r) \left(\delta_{jk} r^2 - r_j r_k\right) \qquad (1)$$

- Identify how components of tensor of inertia transform under rotations of coordinates

$$r_i \to r_i = \sum_{i'} \mathcal{R}_{ii'} r_{i'}$$

As You may remember, in case of rotations $\mathcal{R}_{ij}$ are orthogonal matrices, e.g:

$$\hat{\mathcal{R}}_Z = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \hat{\mathcal{R}}_Z \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x\cos\theta + y\sin\theta \\ -x\sin\theta + y\cos\theta \\ z \end{pmatrix}$$

The idea is to express I_{ij} in different frames (e.g. lab frame and body's rest frame) to be able to relate them (and thus simplify analysis)

Tensor of inertia: transformation of components

$$I_{jk} = \int dV\, \rho(r) \left(\delta_{jk} r^2 - r_j r_k\right) \qquad (1)$$

– From definition (1): second term gets two $\hat{\mathcal{R}}$-matrices, one for each "free" index

$$r_j r_k \to \sum_{j' k'} \mathcal{R}_{jj'} \mathcal{R}_{kk'} r_{j'} r_{k'}$$

so it "looks like" that I_{jk} should transform similarly as

$$I_{jk} \to \sum_{j' k'} \mathcal{R}_{jj'} \mathcal{R}_{kk'} I_{j' k'} \qquad (2)$$

–The term $\sim\ r^2 \delta_{ik}$ is invariant (r^2 is absolute value of radius-vector and does not change).

⇒Seems we are in trouble, two terms in (1) transform differently (?)

... but we recall that

$$\sum_{j' k'} \mathcal{R}_{j\, j'} \mathcal{R}_{k\, k'} \delta_{j' k'} = \sum_{j'} \mathcal{R}_{j\, j'} \mathcal{R}_{k\, j'} = \delta_{jk}$$

(orthogonality), so can interpret that it also acquires two $\hat{\mathcal{R}}$-matrices per each index (!)

⇒ The tensor I_{jk} transforms as given in (2).

Tensor of inertia: transformation of components

$$I_{jk} \to \sum_{j'k'} \mathcal{R}_{jj'} \mathcal{R}_{kk'} I_{j'k'} \qquad (1)$$

- This result is very important:

–In any frame

$$I_{jk} = \int dV\, \rho(\mathbf{r},\ t) \left(\delta_{jk} r^2 - r_j r_k\right) =$$
$$= \int dV\, \rho(\mathbf{r},\ t) \begin{pmatrix} y^2 + z^2 & -xy & -xz \\ -xy & x^2 + z^2 & -yz \\ -xz & -yz & x^2 + y^2 \end{pmatrix}$$

– in lab frame the density depends on orientation of the body (=on time), for this reason all components $I_{ij}^{(\mathrm{lab})}$ also depend on time

–in body's rest frame ρ does not depend on time, so $I_{ij}^{(\mathrm{body})}$ are constant

–The equation (1) allows to relate $I_{ij}^{(\mathrm{lab})}$ and $I_{ij}^{(\mathrm{body})}$ if we know orientation of the body (and associated rotation matrices $\mathcal{R}$)

Tensor of inertia - Example

$$I_{jk} = \int dV\, \rho(r) \left(\delta_{jk} r^2 - r_j r_k\right)$$

Evaluate the tensor of inertia for the homogeneous sphere of radius R and mass m.

- The moments of inertia $I_x = I_y = I_z = \frac{2}{5} m R^2$, which by definition coincide with components I_{xx}, I_{yy}, I_{zz}
- For nondiagonal components we have:

$$I_{xy} = -\int \rho dV\, x\, y = -\rho \int_0^R r^2 dr \int_0^\pi \sin\theta d\theta \int_0^{2\pi} d\phi\, r^2 \sin^2\theta \underbrace{\sin\phi\cos\phi}_{\frac{1}{2}\sin 2\phi} = 0$$

$$I_{xz} = -\int \rho dV\, x\, z = -\rho \int_0^R r^2 dr \int_0^\pi \sin\theta d\theta \int_0^{2\pi} d\phi\, r^2 \frac{\sin 2\theta}{2} \cos\phi = 0$$

$$I_{yz} = -\int \rho dV\, y\, z = -\rho \int_0^R r^2 dr \int_0^\pi \sin\theta d\theta \int_0^{2\pi} d\phi\, r^2 \frac{\sin 2\theta}{2} \sin\phi = 0$$

Tensor of inertia - Example

$$\Rightarrow I_{jk} = \frac{2}{5} mR^2 \delta_{jk} = \frac{2}{5} mR^2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Why nondiagonal components vanish?

The sphere has 3 symmetry planes (invariant w.r.t. 3 *independent* transformations $x \to -x$, $y \to -y$, $z \to -z$ in Cartesian coordinates), and nondiagonal components change sign under such transforms, that's why they vanished

Control question 1

Definition:

$$I_{jk} = \int dV\, \rho(r) \left(\delta_{jk} r^2 - r_j r_k \right)$$

Prove that for any object with arbitrary mass density $\rho(\boldsymbol{r})$, the eigenvalues $\lambda_1, \lambda_2, \lambda_3$ of I_{jk} are real-valued, and the eigenvectors which correspond to different eigenvalues are orthogonal to each other.

Prove that in any frame

$$\det\left(\hat{I}\right) = \lambda_1 \lambda_2 \lambda_3, \qquad \mathrm{Tr}\left(\hat{I}\right) = \sum_{j=k} I_{jk} = \sum_j I_{jj} = \lambda_1 + \lambda_2 + \lambda_3$$

Control question (reply)

Solution:

1) The matrix I_{jk} is symmetric and real-valued $\Rightarrow$ It is hermitian
2) For hermitian matrices all eigenvalues are real-valued, and eigenvectors orthogonal to each other, Q.E.D.
3) In the basis where the matrix I_{jk} is NOT diagonal, we may represent it as

$$I_{jk} = \left[R \cdot \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix} \cdot R^{-1} \right]_{jk} = \left[R \cdot I^{(\mathrm{diag})} \cdot R^{-1} \right]_{jk}$$

where R is some rotation matrix which relates two basis, so

$$\det\left[R I^{(\mathrm{diag})} R^{-1} \right] = \cancel{\det\left[R\right]} \det\left[I^{(\mathrm{diag})} \right] \cancel{\det\left[R\right]^{-1}} = \det\left[I^{(\mathrm{diag})} \right] = \lambda_1 \lambda_2 \lambda$$

$$\mathrm{Tr}\left[R I^{(\mathrm{diag})} R^{-1} \right] = \mathrm{Tr}\left[I^{(\mathrm{diag})} \cancel{R^{-1}} \cancel{R} \right] = \mathrm{Tr}\left[I^{(\mathrm{diag})} \right] = \lambda_1 + \lambda_2 + \lambda_3$$

Control question 2

Definition:

$$I_{jk} = \int dV\, \rho(r) \left(\delta_{jk} r^2 - r_j r_k\right)$$

We discussed that under rotations the components of I_{jk} transform as

$$I_{jk} \to \sum_{j'k'} \mathcal{R}_{jj'} \mathcal{R}_{kk'} I_{j'k'}$$

$$\text{where } \mathcal{R}_{ii'} \text{ is the rotation matrix} \qquad r_i \to r_i = \sum_{i'} \mathcal{R}_{ii'} r_{i'}$$

Assume that we found all eigenvalues λ_a and corresponding eigenvectors $\boldsymbol{v}^{(\lambda)} \equiv v_a^{(\lambda)} \boldsymbol{e}_a$ in one reference frame. How these eigenvalues and eigenvectors transform under rotations ???

Control question 2 (solution)

As You may guess the eigenvalues don't change, and vectors can be just rotated with matrix $\mathcal{R}_{ij}$

Lemma: If the matrix I_{jk} has eigenvalues λ_a and corresponding eigenvectors $\boldsymbol{v}^{(\lambda)} \equiv \sum_j v_j^{(\lambda)} \boldsymbol{e}_j \equiv v_1^{(\lambda)} \boldsymbol{e}_1 + v_2^{(\lambda)} \boldsymbol{e}_2 + v_3^{(\lambda)} \boldsymbol{e}_3$, such that

$$\sum_k I_{jk} v_k^{(\lambda_a)} = \lambda_a v_j^{(\lambda_a)}, \qquad a = 1, 2, 3$$

then the matrix $\tilde{I}_{jk} = \sum_{j'k'} \mathcal{R}_{jj'} \mathcal{R}_{kk'} I_{j'k'}$ has eigenvectors $\tilde{\boldsymbol{v}}^{(\lambda)} = \sum_j \tilde{v}_j^{(\lambda)} \boldsymbol{e}_j, \qquad \tilde{v}_j^{(\lambda)} = \sum_k \mathcal{R}_{jk} v_k^{(\lambda)}$ with exactly the same eigenvalues,

$$\sum_k \tilde{I}_{jk} \tilde{v}_k^{(\lambda_a)} = \lambda_a \tilde{v}_k^{(\lambda_a)}, \qquad a = 1, 2, 3$$

Control question 2 (solution)

Proof:

$$\sum_k \tilde{I}_{jk} \tilde{v}_k^{(\lambda_a)} = \sum_k \sum_{j'k'} \mathcal{R}_{jj'} \mathcal{R}_{kk'} I_{j'k'} \sum_\ell \mathcal{R}_{k\ell} v_\ell^{(\lambda_a)} = \sum_{j'k'} \mathcal{R}_{jj'} I_{j'k'} \sum_\ell \delta_{k',\ell} v_\ell^{(\lambda_a)}$$

where at the last step in 1st line we used orthogonality

$$\sum_k \mathcal{R}_{kk'} \mathcal{R}_{k\ell} = \delta_{k',\ell}$$

Now we'll use the fact that $v_k^{(\lambda)}$ are components of eigenvector of I_{jk}:

$$\sum_k \tilde{I}_{jk} \tilde{v}_k^{(\lambda_a)} = \sum_{j'k'} \mathcal{R}_{jj'} I_{j'k'} v_{k'}^{(\lambda_a)} = \sum_{j'k'} \mathcal{R}_{jj'} \lambda_a v_{j'}^{(\lambda_a)} = \lambda_a \tilde{v}_j^{(\lambda_a)}, \quad \text{Q.E.D.}$$

Control question 3

Prove that all eigenvalues $\lambda_1, \lambda_2, \lambda_3$ of I_{jk} are positive numbers just using definition and mathematical properties of I_{jk}. Find how they are related to $\rho(r)$.

Control question 3 (solution)

Eigenvalues don't depend on on reference frame, and can be found in any reference frame

$\Rightarrow$Can choose reference frame K where axes $\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}}$ coincide with eigenvectors $\boldsymbol{v}^{(\lambda)}$ of I_{jk}

$-$In this frame, I_{jk} is diagonal (otherwise, vector $\boldsymbol{v}^{(\lambda)}$ would not be eigenvalue), so

$$I_{jk}^{(K)} = \int dV\, \rho(\boldsymbol{r},\, t) \begin{pmatrix} y^2 + z^2 & 0 & 0 \\ 0 & x^2 + z^2 & 0 \\ 0 & 0 & x^2 + y^2 \end{pmatrix}$$

and eigenvalues of I_{jk} clearly coincide with diagonal elements of $I_{jk}^{(K)}$

Control question 4

Assume that $\lambda_1, \lambda_2, \lambda_3$ are eigenvalues of I_{jk}. Demonstrate that if $\boldsymbol{n}$ is an arbitrary unit vector, then "moment of inertia"

$$I_n = n \cdot \hat{I} \cdot n = n_j n_k I_{jk}$$

is always bound by

$$\lambda_{\min} \leq I_n \leq \lambda_{\max}$$

where $\lambda_{\min}$ and $\lambda_{\max}$ are minimal and maximal values in $\{\lambda_1, \lambda_2, \lambda_3\}$

As we discussed earlier, we may always rewrite

$$\hat{I}_{jk} = \left[R \cdot \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix} \cdot R^T \right]_{jk} = \left[R \cdot I^{(\mathrm{diag})} \cdot R^T \right]_{jk} = R_{ja} R_{kb} I^{(\mathrm{diag})}_{ab}$$

so for the moment of inertia

$$I_n = n_j n_k I_{jk} = n_j n_k R_{ja} R_{kb} I^{(\mathrm{diag})}_{ab}$$

Define: vector $\tilde{\boldsymbol{n}}$ whose components are given by $\tilde{n}_a = R_{ja} n_j = \left(R^T \cdot n \right)$.
Clearly, $\tilde{\boldsymbol{n}}$is a unit vector (result of rotation of $\boldsymbol{n}$), so $\tilde{n}_1^2 + \tilde{n}_2^2 + \tilde{n}_3^2 = 1$

$$I_n = \tilde{n}_a \tilde{n}_b I^{(\mathrm{diag})}_{ab} = \lambda_1 \, \tilde{n}_1^2 + \lambda_2 \, \tilde{n}_2^2 + \lambda_3 \, \tilde{n}_3^2 = \lambda_1 \, \tilde{n}_1^2 + \lambda_2 \, \tilde{n}_2^2 + \lambda_3 \, \left(1 - \tilde{n}_1^2 - \tilde{n}_2^2 \right) =$$

$$= \left(\lambda_1 - \lambda_3 \right) \, \tilde{n}_1^2 + \left(\lambda_2 - \lambda_3 \right) \, \tilde{n}_2^2 + \lambda_3$$

Similarly, we may eliminate $\tilde{n}_1$ and rewrite it as:

$$I_n = \left(\lambda_3 - \lambda_1 \right) \, \tilde{n}_3^2 + \left(\lambda_2 - \lambda_1 \right) \, \tilde{n}_2^2 + \lambda_1$$

Control question 4 (solution)

$$I_n = (\lambda_1 - \lambda_3)\ \tilde{n}_1^2 + (\lambda_2 - \lambda_3)\ \tilde{n}_2^2 + \lambda_3 = \tag{1}$$

$$= (\lambda_3 - \lambda_1)\ \tilde{n}_3^2 + (\lambda_2 - \lambda_1)\ \tilde{n}_2^2 + \lambda_1 \tag{2}$$

Assume for definiteness that $\lambda_{\max} = \lambda_1$ and $\lambda_{\min} = \lambda_3$.

$-$Since $\lambda_1 - \lambda_3 \geq 0$, $\lambda_2 - \lambda_3 \geq 0$, conclude from (1) that

$$I_n - \lambda_3 = (\lambda_1 - \lambda_3)\ \tilde{n}_1^2 + (\lambda_2 - \lambda_3)\ \tilde{n}_2^2 \geq 0,$$

i.e. $I_n \geq \lambda_3 = \lambda_{\min}$

$-$Since $\lambda_3 - \lambda_1 \leq 0$, $\lambda_2 - \lambda_1 \leq 0$, conclude from (2) that

$$I_n - \lambda_1 = (\lambda_3 - \lambda_1)\ \tilde{n}_3^2 + (\lambda_2 - \lambda_1)\ \tilde{n}_2^2 \leq 0 \tag{3}$$

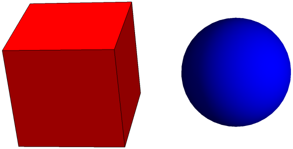
i.e. $I_n \leq \lambda_1 = \lambda_{\max}$

Tensor of inertia - Classification

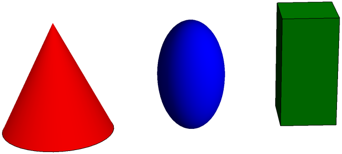
►Since I_{jk} is symmetric, real-valued, can always diagonalize it

−Physically three different scenarios:

- $I_1 = I_2 = I_3$-spherical top (spherically symmetric)



- $I_1 = I_2 \neq I_3$-symmetric top (axially symmetric)



- $I_1 \neq I_2 \neq I_3$-asymmetric top

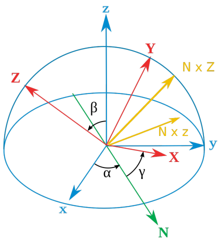
−whatever object without symmetries

Description in lagrangian formalism

- Now we'll try to formulate and solve the problem of motion of a rigid body in lagrangian formalism
  - First we'll introduce Euler's angles which describe finite rotations
  - We'll work in the body's rest frame, since $I_{ij} = \mathrm{const}$ there
  - Next we'll construct the Lagrangian and try to solve equations of motion (using possible integrals of motion)

Angular velocity

Assume that the orientation of rigid body is described in terms of Euler angles α, β, γ. Evaluate the components of vector $\vec{\Omega}$ in the rest frame of the body



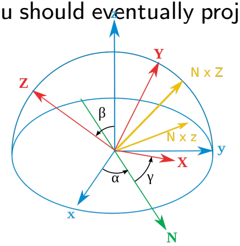
Description in lagrangian formalism

- It is convenient to decompose the angular vector $\boldsymbol{\Omega}$ as

$$\boldsymbol{\Omega} = \dot{\alpha}\boldsymbol{e}_\alpha + \dot{\beta}\boldsymbol{e}_\beta + \dot{\gamma}\boldsymbol{e}_\gamma$$

where $\boldsymbol{e}_\alpha$, $\boldsymbol{e}_\beta$, $\boldsymbol{e}_\gamma$ are unit vectors in directions of corresponding rotation axes:
▷ $\boldsymbol{e}_\alpha$ points in vertical direction of old axis Z (blue arrow)
▷$\boldsymbol{e}_\beta$ points in direction of line of nodes N (green arrow)
▷$\boldsymbol{e}_\gamma$ points in direction of line of new axis Z (red arrow)

- You should eventually project vector $\boldsymbol{\Omega}$ onto the axes X, Y, Z of the body rest frame

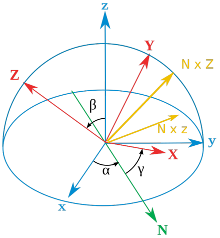


Vector $\boldsymbol{\Omega}$ in the body rest frame:

$$\Omega_1 = \dot{\alpha}\sin\beta\sin\gamma + \dot{\beta}\cos\gamma$$
$$\Omega_2 = \dot{\alpha}\sin\beta\cos\gamma - \dot{\beta}\sin\gamma$$
$$\Omega_3 = \dot{\alpha}\cos\beta + \dot{\gamma}.$$

(can get the result making projections geometrically)

Description in lagrangian formalism



Vector $\vec{\Omega}$ in the body rest frame:

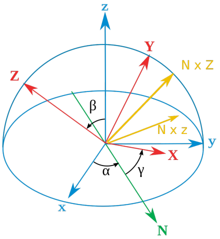
$$\Omega_1 = \dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma$$
$$\Omega_2 = \dot{\alpha} \sin \beta \cos \gamma - \dot{\beta} \sin \gamma$$
$$\Omega_3 = \dot{\alpha} \cos \beta + \dot{\gamma}.$$

Kinetic energy

$$T = \frac{I_1 \Omega_1^2}{2} + \frac{I_2 \Omega_2^2}{2} + \frac{I_3 \Omega_3^2}{2}$$

$-$direct substitution gives result (lengthy but simple by structure)

Description in lagrangian formalism



Vector $\vec{\Omega}$ in the body rest frame:

$$\Omega_1 = \dot{\alpha}\sin\beta\sin\gamma + \dot{\beta}\cos\gamma$$
$$\Omega_2 = \dot{\alpha}\sin\beta\cos\gamma - \dot{\beta}\sin\gamma$$
$$\Omega_3 = \dot{\alpha}\cos\beta + \dot{\gamma}.$$

Symmetric top

Write out the kinetic energy of the symmetric top ($I_1 = I_2 \neq I_3$) whose orientation is described by Euler angles α, β, γ. Write out the Lagrangian for the case of free motion and integrate the EOM.

Description in lagrangian formalism

Symmetric top:

▶Kinetic energy and lagangian:

$$
\begin{aligned}
T &= \frac{I_1}{2}\left(\dot{\alpha}^2 \sin^2\beta + \dot{\beta}^2\right) + \frac{I_3}{2}\left(\dot{\alpha}\cos\beta + \dot{\gamma}\right)^2 \\
U &= 0 \Rightarrow T = E = L = \mathrm{const}
\end{aligned}
$$

Generalized momenta ($p_\alpha = \partial L/\partial \dot{q}_a$):

$$
\begin{aligned}
M_\beta &= I_1\dot{\beta} \neq \mathrm{const}\,(\beta\ \text{ not cyclical}) \\
M_\alpha &= \dot{\alpha}\left(I_1\sin^2\beta + I_3\cos^2\beta\right) + I_3\dot{\gamma}\cos\beta = \mathrm{const} \\
M_\gamma &= I_3\left(\dot{\alpha}\cos\beta + \dot{\gamma}\right) = \mathrm{const.}
\end{aligned}
$$

▶Note that $\boldsymbol{M} = \mathrm{const}$ and in lab frame all 3 components M_x, M_y, M_x are conserved.

▶ In moving frame directions of basis vectors $\boldsymbol{e}_\alpha$, $\boldsymbol{e}_\beta$, $\boldsymbol{e}_\gamma$ change, so projections of $\boldsymbol{M}$ onto these axes are NOT constants

$$
\dot{\alpha} = \frac{M_\alpha - M_\gamma\cos\beta}{\tilde{I}_1\sin^2\beta}, \quad \dot{\gamma} = \frac{M_\gamma}{I_3} - \cos\beta\frac{M_\alpha - M_\gamma\cos\beta}{\tilde{I}_1\sin^2\beta}.
$$

Description in lagrangian formalism

$$\dot\alpha = \frac{M_\alpha - M_\gamma\cos\beta}{\tilde I_1 \sin^2\beta}, \quad \dot\gamma = \frac{M_\gamma}{I_3} - \cos\beta\frac{M_\alpha - M_\gamma\cos\beta}{\tilde I_1\sin^2\beta}.$$

$$\Rightarrow E = \frac{M_\gamma^2}{2I_3} + \frac{I_1}{2}\dot\beta^2 + \frac{(M_\alpha - M_\gamma\cos\beta)^2}{2I_1\sin^2\beta}.$$

$$\lambda = \cos\beta; \quad \dot\lambda = -\dot\beta\sin\beta,$$

$$\dot\beta = -\frac{\lambda}{\sqrt{1-\lambda^2}}.$$

$$\frac{I_1}{2}\dot\lambda^2 + \lambda^2\underbrace{\left(E + \frac{M_\gamma^2}{2}\left(\frac{1}{I_1} - \frac{1}{I_3}\right)\right)}_{a/2} -$$

$$-\underbrace{\frac{M_\alpha M_\gamma}{I_1}}_{b}\lambda + \underbrace{\left(\frac{M_\alpha^2}{2I_1} + \frac{M_\gamma^2}{2I_3} - E\right)}_{c/2} = 0$$

Description in lagrangian formalism

$$\Rightarrow E = \frac{M_\gamma^2}{2I_3} + \frac{I_1}{2}\dot{\beta}^2 + \frac{(M_\alpha - M_\gamma \cos\beta)^2}{2I_1 \sin^2\beta}.$$

$$\lambda \quad = \cos\beta,$$

$$\frac{I_1}{2}\dot{\lambda}^2 + \lambda^2 \frac{a}{2} - b\lambda + \frac{c}{2} = 0$$

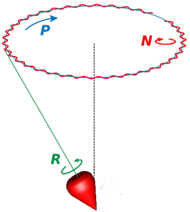
►Introduce $\mu(t) := \lambda(t) - \frac{b}{a}$, can get

$$\frac{I_1}{2}\dot{\mu}^2 + \frac{a}{2}\mu^2 = \underbrace{\frac{b^2}{2a} - \frac{c}{2}}_{=\mathcal{E}}$$

$\Rightarrow$coincides with expression for harmonic oscillator with "mass" I_1, "elasticity" a and energy $\mathcal{E}$

$$\Rightarrow \qquad \mu(t) = A\cos(\omega t + \varphi_0),$$
$$A = \sqrt{2\mathcal{E}/a}, \quad \omega = \sqrt{a/I_1}$$

$$\Rightarrow \cos\beta = \frac{b}{a} + A\cos(\omega t + \varphi_0)$$



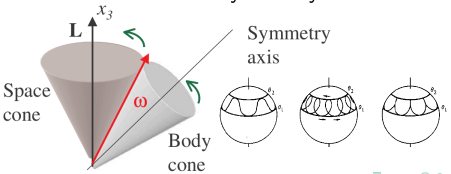
oscillations with frequency

$$\omega^2 = \frac{a}{I_1},$$

around "center of equilibrium"

$$\lambda_0 \equiv \cos\beta_0 = \frac{b}{a},$$

- Visualization of the symmetry axis motion



Summary

- Rigid body = system with 6 degrees of freedom
- Transformation of coordinates from body's rest frame to lab frame:

$$r_a^{(\mathrm{lab})}(t) = R_{ab}\left(\psi(t),\, \theta(t),\, \varphi(t)\right) r_b^{(\mathrm{body})} + X_a(t), \quad a = 1, 2, 3 \;\; (a = x, y, z)$$

- Kinetic energy can be represented as energy of the center of mass and rotation

$$T = \frac{M\dot{\boldsymbol{R}}_{\mathrm{CM}}^2}{2} + \frac{1}{2} I_{jk}\Omega_j\Omega_k$$

  - Tensor of inertia $I_{jk} = \int dV\, \rho(r)\left(\delta_{jk}r^2 - r_j r_k\right)$ is constant in the body's rest frame
  - in the lab frame gets time dependence; may be related to body's rest frame via

$$I_{jk}^{(\mathrm{lab})} = \sum_{j'k'} R_{jj'}(t) R_{kk'}(t) I_{j'k'}^{(\mathrm{body})}, \quad R_{ij} = \text{rotation matrix}$$

  - Components of the angular velocity Ω in the body's rest frame (Euler's ZXZ convention)

$$\boldsymbol{\Omega} = \left\{\dot{\alpha}\sin\beta\sin\gamma + \dot{\beta}\cos\gamma,\, \dot{\alpha}\sin\beta\cos\gamma - \dot{\beta}\sin\gamma,\, \dot{\alpha}\cos\beta + \dot{\gamma}\right\}$$